Interaction Obstructions, Resonant PT Breaking, and Doubled Jordan Symmetry in Fourth-Order Theories

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Abstract

We formulate a deformation problem for the canonical positive pseudo-Hermitian metric of the Pais–Uhlenbeck oscillator and for its Krein/ghost-parity alternative, and follow both into interacting fourth-order theories. Four results organize the paper. (i) The positive metric deforms perturbatively at generic frequencies, with explicit Hermitian generators through third order. (ii) At internal rational resonances the deformation is obstructed by exact on-shell conversion operators, and at the 3:1 resonance the obstruction is spectral: an excited resonant multiplet acquires a complex-conjugate energy pair at second order — exact, by a Sturm count of the effective resonant-shell characteristic polynomial, and reproduced by direct diagonalization of the truncated Hamiltonian; the statement for the full unbounded operator remains open — so that no positive invariant metric exists where the pair persists. (iii) In the perfect-square field theory the momentum continuum turns such isolated resonances into generic scattering shells: the branch-changing process $H + L \to L + L$ obstructs the analytic second-order deformation of the pointed-sector positive metric on an open subset of its shell, with an exact value at a rational kinematic point. (iv) In the two-field frame $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ — the $O(1,1)$ embedding of Bateman and Turok, whose exchange symmetry $U \leftrightarrow V$ they identified as ghost parity — the exchange maps two oppositely oriented null Jordan sectors into one another, its linearization is the bounded confluent limit of the regulated branch parity on the doubled space, and the obstruction of (iii) sits entirely in the exchange-odd component of the on-shell T, which is exactly ghost-parity pseudo-Hermitian. A self-contained charge-null lemma and regulated embedding further show that one-sided charge nullity is an enhanced boundary property, not a property of the split theory: the mapped vacuum obeys $S_{UU}/S_{VV} = (\delta/2g)^2 = \epsilon/g$ and becomes one-sided only on the confluent line, with the Bateman–Turok massless theory at $\epsilon = \mu^2 = 0$. Exact split-shell Krein pseudo-unitarity and boundary one-sidedness are therefore distinct; the two completions are inequivalent in the interacting theory. All finite-dimensional and modewise identities are machine-verified; derivations of the principal results are given in the appendices. An independent Science Forge computation subsequently extends the discrete resonance hierarchy to order six: the 5:3 and 7:1 loci vanish through orders two–five and obstruct at order six on exactly the predicted conversion monomials. Those two instances are imported here as externally reproduced exact evidence; the all-coprime hierarchy remains a conjecture. A separate exact `REDUCED-MODE` continuation of the scalar loop programme now completes the final-state collinear logarithmic response. It finds that the real finite-part normalization changes by

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$3\lambda^6 \log c/(512\pi^4 s)$, whereas every continuous axis-compatible parent/daughter regulator gluing, including the physical pair threshold, has zero constant virtual response. This obstructs the ordinary independent-mass infrared prescription on that carrier; it is not a full NLO probability, failure of dressed-state or resummed constructions, or a gravitational result. An exact finite-carrier successor changes the measured object. Correctly normalizing its dimensionless projector by the Born rate gives Gram coefficient $1/48$ per pair, amplitude $\sqrt{3}/12$, and hard/collinear blocks $-1/16$ and $+1/16$; multiplying by the Born rate cancels the physical $3/512$ response. The ordinary single-denominator cubic Fock generator is, however, exactly zero in the massless collinear limit. The remaining gate is now narrower: expanding the published composite map gives the exact order-λ two-annihilator kernel, and all of its apparent t, t^2 Jordan growth cancels on the two-field carrier. Its Krein Gram has cubic splitting-endpoint poles, however, so the distributional continuum projector and its dynamically derived $1/48$ coefficient are not yet constructed. An exact canonical-algebra countermodel further shows that CCR preservation, projector idempotence, trace preservation, and exchange symmetry alone retain three endpoint parameters. Thus $1/48$ is compatible but not selected without the deferred continuum transport or another independently justified physical endpoint condition. Independently, the projected three-channel hard loop logarithm closes the Callan–Symanzik equation exactly with the Born rate and one-loop beta function. This yields the positive leading-log nonforward prediction $d\sigma_{\text{hard}}/d\Omega = 24\pi^2/[25 s \log^2(s/\Lambda^2)]$. It is a physical hard/window baseline, not a completed inclusive NLO probability; the collinear endpoint construction remains open.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

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1 Introduction

A companion series [11, 12, 13, 14] analyzed the *free* fourth-order theories — the Pais–Uhlenbeck (PU) oscillator, the fourth-order scalar field, and linearized quadratic gravity — and established a three-level structure: one complex spectral covariance, several real forms (involutions), and several completions. The positive pseudo-Hermitian completion of Bender and Mannheim [1, 2] and the Krein/ghost-parity completion of Bateman and Turok [4] are real forms of one free theory, distinguished at Jordan degenerations. This paper begins the interacting classification: which of these free structures can be continuously deformed when nonlinear processes are switched on?

The organizing mechanism, established here first for discrete oscillator resonances and then for continuum scattering shells, is

$$\text{metric deformation} \longrightarrow \text{on-shell cohomological obstruction} \longrightarrow \text{spectral complexification.}$$

Results. For the PU oscillator with the cubic vertex $V = -iy^3$ inherited from a real z^3 interaction: the deformation exists at first order for all $\omega_1 > \omega_2 > 0$ and at second and third order for generic frequencies (Sections 3–4); it is obstructed at second order at $\omega_1 = 3\omega_2$ by the exact on-shell conversion class $\propto a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3$, gauge-independently (Section 5); the obstruction is spectral — a complex-conjugate pair appears in the $E = 27\omega_2$ multiplet at $O(\zeta^2)$, certified exactly (Section 6); a third-order obstruction appears at $\omega_1 = \frac{3}{2}\omega_2$, and a transfer-lattice selection rule organizes the hierarchy (Section 7), whose first prediction is confirmed: 5:1 is unobstructed through order 3 and obstructed at order 4, exactly on the predicted monomial (Proposition 7.4); independent exact order-six calculations find the predicted first obstructions at 5:3 and 7:1 (Proposition 7.5) — the hierarchy is thereby sampled at five qualitatively distinct resonances through order six; the deformation diverges toward the Jordan boundary, with exact leading asymptotics at first order and measured geometric growth beyond (Section 8).

For the perfect-square field theory $S = -\frac{1}{2}\int(\Box\phi + \lambda(\partial\phi)^2)^2$, regulated by a mass splitting: the cubic vertex is exactly the Källén polynomial (Section 9); an even-ghost selection rule protects the positive metric at first order even above the ghost-decay threshold, while the sectorwise branch parity is obstructed there by the real decay (Section 10); at second order the branch-changing scattering $H + L \to L + L$ obstructs the analytic deformation of the pointed-sector positive metric on an open subset of its shell, with the exact value $401\sqrt{6}/39424$ at a rational kinematic point in fully specified conventions (Section 12); the exact two-field rewriting exhibits the extended theory's exchange symmetry, its two null Jordan sectors, and the interaction-generated Jordan dynamics (Section 11); and the regulated branch parity, singular on one confluent chain, converges on the doubled space to the linearization of that exchange (Section 13).

Relation to prior work. The two-field $O(1,1)$ embedding, the identification of the exchange $U \leftrightarrow V$ as ghost parity and internal charge conjugation, the quantum embedding relating the two-field and fourth-order Hamiltonians and S-matrices, and a Krein-space Born rule with positivity through weak ghost symmetry are all due to Bateman and Turok [4]; none of that structure is claimed here. Likewise standard are: complex eigenvalues of pseudo-Hermitian operators occurring in conjugate pairs with matching Jordan data [6], the forced indefiniteness of any compatible metric in their presence [7], antilinear (CPT-type) symmetry as the origin of real-or-conjugate-pair spectra [8], complex-ghost-pair unitarity of Lee–Wick type [9] (physically distinct: there the complex ghosts are excluded from asymptotic states), and positive invariant graph subspaces of block J-self-adjoint operators [10]. What this paper adds is the *deformation-obstruction theory of the positive pseudo-Hermitian metric* in the interacting theories: the obstruction complex itself, the exact 3:1 and 3:2 conversion classes and the transfer-lattice selection rule, the certified spectral pair and its broken-PT tongue, the continuum scattering-shell obstruction, the confluent bridge from the split branch parity to the doubled Jordan exchange, and the resulting separation statement: the Bateman–Turok Krein completion is *not* equivalent to a hidden positive pseudo-Hermitian completion, because the latter is obstructed by on-shell branch-changing interactions while the doubled Krein pairing remains exactly pseudo-unitary on the verified matrix elements (Proposition 12.4). The stronger one-sided-charge condition entering the Bateman–Turok Born rule is instead a boundary property (Proposition 14.2); its explicit Born-trace evaluation on a fixed split-mass shell has since been completed and shown not to test the charge-null mechanism, because the truncated shell carries no boost action. The first complete final-state collinear logarithmic audit is instead obstructed by the ordinary axis-compatible independent-mass prescription (Remark 14.3). A beyond-tree quotient trace therefore requires a new infrared/asymptotic-state architecture, not another fixed-shell evaluation.

Strength of the statements. Throughout we distinguish (a) *analytic metric obstruction* — nonexistence of an order-by-order deformation of the canonical metric in the specified representation and operator class — from (b) *absolute nonexistence of positive metrics.* For the oscillator at 3:1 the stronger statement (b) holds wherever the certified complex pair persists, since a positive metric would make the Hamiltonian similar to a self-adjoint operator; the qualifications (formal perturbation theory, unbounded cubic interaction, truncation) are stated where they apply. For the field theory only (a) is established here. "Generic" is likewise used in three distinct senses, always resolvable from context: *generic oscillator frequencies* (outside the finite resonance-candidate set at the stated order), *generic field obstruction* (nonzero on a nonempty open subset of an allowed scattering shell), and *kinematically open for every mass pair* (existence of the shell, which does not by itself imply nonvanishing of the obstruction at every point of it). The status ledger is: R_1 Jordan scaling — theorem; R_2, R_3 scaling — computational propositions; $R_n \sim \epsilon^{-3n/2}$ — conjecture; the 5:1 obstruction — computational proposition (confirmed at order 4, Proposition 7.4); the 5:3 and 7:1 order-six obstructions — clean-pinned externally reproduced computational proposition (Proposition 7.5); a full-field complex spectrum — open; the exact doubled classical involution — theorem (Bateman–Turok, restated); its quantum implementation — open; the κ-odd localization of the shell obstruction — computational proposition at the rational point (Proposition 12.4); one-sidedness of the regulated vacuum image — computational proposition, exact iff $\epsilon = 0$ in the stated charge frame (Proposition 14.2); the boundary Born-trace evaluation — open, with the fixed-shell route excluded rather than merely unattempted because the charge-null lemma's hypothesis is absent there (Section 14); the complete final-state collinear logarithmic response on the ordinary axis-compatible independent-mass regulator class — exact `REDUCED-MODE` obstruction (Remark 14.3); existence of a normalized finite coherent cross-multiplicity projector at the residual coefficient — exact `LOCAL-ALGEBRAIC` plus `REDUCED-MODE` classification (Remark 14.4); derivation of that transport from BT dynamics, a continuum dressed projector, full NLO quotient trace, and gravitational lift of that infrared statement — open.

Bookkeeping. The deformation order is counted by a formal parameter ζ: $H(\zeta) = H_0 + \zeta H_3 + \zeta^2 H_4$, defining $R_1, R_2, \ldots$ and the obstruction orders. The perfect-square coupling λ is a fixed parameter: in the two-field frame the quadratic Jordan structure and the vertex coefficients carry $g = \lambda^2$, while the nonlinear field map itself depends on λ. In the oscillator sections ζ multiplies the cubic vertex and coincides with the coupling.

Verification and reproducibility. Every finite-dimensional, modewise, and distributional identity in this paper is machine-verified at the root of the `area9innovation/weyl-gravity` repository accompanying the series (SymPy 1.14.0, NumPy 2.4.4, Python 3.14; cite the repository commit hash to fix the artifact). Appendix G maps checks to theorems and classifies each check as exact-symbolic, exact-rational (Sturm/charpoly), or high-precision numerical. Appendices A–F contain readable derivations of the principal results. Operator-domain caveats (unbounded interactions, formal series) are flagged where they occur. The order-six extension is pinned separately by `bridge/science_forge/certificates/SCIENCE_FORGE_PU_ORDER6_IMPORT.json` to Tango commit `ccee11d08de28d3e2681db209264887950666e87`. It records exact agreement between two Forge backends and an independent SymPy recomputation, while also recording that the upstream certificate was emitted by a dirty development toolchain. A second run built the compiler from a clean `git archive` of that commit and reproduced all 31 checks and all three SymPy golden comparisons; the clean replay receipt is part of the import. The scalar real–virtual continuation is pinned under `reverse_physics/certificates/` by certificate `REVERSE_PHYSICS_BT_REAL_VIRTUAL_AXIS_GLUING_V1.json` at standalone-repository commit `0daa9ed5507b983322b6e04cccf78907edf482b5`. Its producer and method-distinct invariant-Källén verifier use exact arithmetic and run sequentially under a 500,000 KB virtual-memory cap; the certificate preserves the `REDUCED-MODE` dependency boundary and explicitly leaves the full NLO quotient trace uncomputed. The finite coherent predecessor `REVERSE_PHYSICS_BT_COLLINEAR_ PROJECTOR_TRANSPORT_V1.json` is retained but marked `SUPERSEDED_NORMALIZATION`: it used an absolute rate coefficient in a dimensionless projector. The correction is pinned by `REVERSE_PHYSICS_BT_ASYMPTOTIC_GENERATOR_PREFLIGHT_V1.json`. Its exact $\mathbb{Q}(\sqrt{3})$ producer and method-distinct verifier reconstruct the Born normalization, projector equations, and cubic collinear scaling. The certificate labels the Jordan asymptotic generator and continuum projector `NOT_CONSTRUCTED`. Its successor `REVERSE_PHYSICS_BT_RT_ JORDAN_KERNEL_V1.json` crosses the first of those gates. It detects an internal oscillator-label inconsistency in Appendix C of Ref. [4], declares the minimal label repair used, derives the nonlinear coefficient directly from the published composite fields, and independently verifies exact secular cancellation. The same certificate fails closed at the nonintegrable endpoint Gram rather than promoting the formal finite-mode generator to a continuum projector. The hard result is pinned by `REVERSE_PHYSICS_BT_UV_HARD_SCATTERING_LAW_V1.json`. It imports the content-addressed Born, beta-function, and projected-hard-log certificates, verifies their Callan–Symanzik identity by an independent rail, and keeps the full inclusive NLO probability `NOT_ESTABLISHED`.

Conventions. Oscillator conventions follow [11, 12]: normalized coordinates $V = (x, y, p, q)$, mode $1 = (x, p)$ at frequency ω_1, mode $2 = (y, q)$ at ω_2, positive normal form $h_0 = \frac{1}{2}[\omega_1\omega_2 p^2 + (\omega_1/\omega_2)x^2 + (\omega_2/\omega_1)q^2 + \omega_1\omega_2 y^2]$, rapidity $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$, $c = \cosh\frac{r}{2} = \omega_1/\sigma$, $s = \sinh\frac{r}{2} = \omega_2/\sigma$, $\sigma = \sqrt{\omega_1^2 - \omega_2^2}$. Operator computations use exact Weyl-symbol Moyal calculus (Appendix A).

2 The deformation complex

Definition 2.1 (Deformation problem and obstruction classes)**.** Let ρ_0 be the canonical free Dyson map, $h_0 = \rho_0 H_0 \rho_0^{-1} = h_0^\dagger$, and $v = \rho_0 V \rho_0^{-1}$ the transported vertex. Seek $\rho_\zeta = \exp[-\frac{1}{2}\sum_n \zeta^n R_n]\rho_0$ with $R_n = R_n^\dagger$ such that $h_\zeta = \rho_\zeta H_\zeta \rho_\zeta^{-1}$ is Hermitian order by order. The first two equations are

$$[h_0, R_1] = v^\dagger - v, \qquad [h_0, R_2] = (v_2^\dagger - v_2) + \tfrac{1}{2}[R_1,\, v + v^\dagger], \tag{1}$$

where v_2 is the transported $O(\zeta^2)$ vertex when present (derivation: Appendix A). For *discrete* spectrum, $h_0 = \sum_E E\, P_E$, the equation $[h_0, R] = S$ has a Hermitian solution iff $P_E S P_E = 0$ for every E; the obstruction class is $\mathfrak{o}_+ = \Pi_{\ker \mathrm{ad}_{h_0}} S = \sum_E P_E S P_E$, and the solution is $R = \sum_{E\neq E'} P_E S P_{E'}/(E - E')$ plus an arbitrary Hermitian element of the commutant. The parallel Krein problem deforms the ghost parity: $[H_0, K_1] = [\kappa_0, V]$, $\{\kappa_0, K_1\} = 0$, with $\mathfrak{o}_K = \Pi_{\ker \mathrm{ad}_{H_0}}[\kappa_0, V]$.

Remark 2.2 (Discrete versus continuous spectrum)*.* The kernel projection above is defined for point spectrum. All field-theoretic computations in this paper are performed at *finite volume* (discrete momenta), where ad_{h_0} has point spectrum and Definition 2.1 applies verbatim; continuum statements are then statements about the finite-volume family, with shell conditions realized by tuning masses to lattice momenta. A distributional (rigged-Fock or spectral-measure) formulation of the continuum obstruction is deferred.

Anti-Hermitian (unitary) generator components are free and do not affect the Hermiticity conditions; commutant additions to R_n are the metric ambiguity. *Gauge independence* of an obstruction class means invariance under both, at all lower orders.

3 The cubic PU oscillator at first order

The real interaction z^3 continues, under the PU rotation $z = iy$, to $V = -iy^3$; the transported vertex is $v = -i(cy + isp)^3$, and $v^\dagger - v = 2ic\,(c^2y^3 - 3s^2yp^2)$.

Theorem 3.1 (First-order deformation)**.** *For every $\omega_1 > \omega_2 > 0$ the first-order equation is solved by the Hermitian Weyl-ordered cubic*

$$R_1 = \mathcal{W}\left[\frac{2c^3}{\omega_1\omega_2}qy^2 + \frac{4c^3}{3\omega_1^3\omega_2}q^3 - \frac{6cs^2(2\omega_1^2-\omega_2^2)}{\omega_1\omega_2(4\omega_1^2-\omega_2^2)}p^2q + \frac{12cs^2\omega_1}{\omega_2(4\omega_1^2-\omega_2^2)}pxy - \frac{12cs^2\omega_1}{\omega_2^3(4\omega_1^2-\omega_2^2)}qx^2\right], \tag{2}$$

with no singular denominator in the physical region; the equivalent Hermitian interaction is $h_1 = \frac{1}{2}(v + v^\dagger) = s(3c^2y^2p - s^2p^3)$. (Verification: substitute (2) into $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$, exact for quadratic h_0.)

4 Second order: generic existence

Theorem 4.1 (Generic second-order deformation)**.** *For incommensurate frequencies the second-order obstruction $\Pi_{\ker \mathrm{ad}_{h_0}} \frac{1}{2}[R_1, v+v^\dagger]$ vanishes; the solved R_2 is Hermitian, and the assembled $h_\zeta = e^{-X/2}\star(h_0+\zeta v)\star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2$, is Hermitian through $O(\zeta^2)$, verified end to end in exact Moyal calculus (including the $\frac{1}{8}[[h_0, R_1], R_1]$ term and the Λ^3 contribution). R_2 carries the resonance denominators $(\omega_1 - \omega_2)$ and $(\omega_1 - 3\omega_2)$; no others can vanish for $\omega_1 > \omega_2 > 0$. (Monomial system: Appendix A.)*

5 The exact 3:1 obstruction

Theorem 5.1 (Second-order obstruction at 3:1)**.** *At $\omega_1 = 3\omega_2$ the first-order deformation still exists, but the second-order class is nonzero and exact:*

$$\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\,\omega_2^4}\left(a_1a_2^{\dagger 3} - a_1^\dagger a_2^3\right), \tag{3}$$

the anti-Hermitian on-shell conversion of one mode-1 quantum into three mode-2 quanta. The class is gauge independent under the full relevant first-order freedom: additions to R_1 of Hermitian commutant elements including the quadratic invariants N_1, N_2, N_1N_2 and the resonant quartic operators *$a_1a_2^{\dagger 3} + a_1^\dagger a_2^3$ and $i(a_1a_2^{\dagger 3} - a_1^\dagger a_2^3)$, and anti-Hermitian first-order generators: $\Pi_{\ker \mathrm{ad}_{h_0}}[G, v+v^\dagger] = \Pi_{\ker \mathrm{ad}_{h_0}}[v - v^\dagger, G] = 0$ for all such G. (Second-order kernel additions to R_2 commute with h_0 and cannot affect the second-order class; they enter first at third order.) Derivation: Appendix B.*

There is thus no analytic second-order deformation of the canonical positive metric at 3:1, within the Weyl-algebra class, continuously connected to the free metric.

6 Resonant PT breaking

Proposition 6.1 (The antilinear symmetry)**.** *Let $\mathcal{A} = \Pi \circ \Theta$, where Θ is complex conjugation in the (x, y)-Schrödinger representation ($x, y \mapsto x, y$, $p, q \mapsto -p, -q$, $i \mapsto -i$) and Π the total parity $(x, y, p, q) \mapsto -(x, y, p, q)$. (The symbol Θ is reserved for this conjugation; it is distinct from the Krein parity generators K_n, the effective spectral matrix $K^{(2)}$, and the momentum cutoff $K_{\max}$.) Then $\mathcal{A}$ is antiunitary, $\mathcal{A}^2 = 1$, and*

$$[\mathcal{A}, h_0] = 0, \qquad [\mathcal{A}, v] = 0, \qquad \textit{hence} \qquad [\mathcal{A}, h_\zeta] = 0 \quad (\zeta \in \mathbb{R}) : \tag{4}$$

h_0 is even and Θ-real; v is odd under both Π and Θ separately ($\Theta v \Theta = -v$, $\Pi v \Pi = -v$), hence invariant under their composition (machine-verified). "PT breaking" below means the appearance of eigenvalues not fixed by this antilinear symmetry, i.e. complex-conjugate pairs.

Theorem 6.2 (Certified complex pair in the $E = 27\omega_2$ multiplet)**.** *At $\omega_1 = 3\omega_2$ ($\omega_2 = 1$), the second-order effective matrix on the ten-dimensional shell $\{|j, 27 - 3j\rangle\}_{j=0}^{9}$ is real tridiagonal with antisymmetric off-diagonal $K_{j,j+1} = -\frac{27\sqrt{3}}{640}\sqrt{(j+1)n(n-1)(n-2)}$, $n = 27 - 3j$, and explicit rational diagonal (Appendix C). Because only the products $K_{j,j+1}K_{j+1,j}$ enter, its characteristic polynomial has* rational *coefficients; an exact Sturm count shows it has exactly eight real roots, hence exactly one complex-conjugate pair, with certified thirty-digit enclosure*

$$\kappa_\pm = 2.448807382199383966\ldots \pm 0.224586593808125108\ldots\, i. \tag{5}$$

Exact diagonalization of the truncated $h_0 + \zeta v$ at $\zeta = 0.02$ (two cutoffs) exhibits the pair with $\operatorname{Im} E = \pm 0.2246\,\zeta^2$ to 0.5%. The lowest resonant doublet $\{|1, 0\rangle, |0, 3\rangle\}$ remains real.

Corollary 6.3. *Wherever the complex pair persists in the interacting spectrum, no positive-definite invariant metric exists — analytic in ζ or not. The hierarchy of claims is: (1) the complex pair in the second-order effective resonant-shell matrix is* exact *(Sturm certificate); (2) its persistence in the interacting truncations is supported by cutoff-stable direct diagonalization; (3) an operator-theoretic statement for the full unbounded cubic Hamiltonian remains open.*

Proposition 6.4 (Broken-PT tongues of the effective multiplet)**.** *Within the second-order effective multiplet, detuning $\delta_r = \omega_1 - 3\omega_2$ adds $j\,\delta_r$ to the shell diagonal, and the spectrum is complex for $|\delta_r| < \delta_c(\zeta) = 38.4151\ldots \times \zeta^2 + o(\zeta^2)$. That an order-$n$ resonance produces an $O(\zeta^n)$ tongue is the corresponding general perturbative expectation, not a proved proposition.*

7 Third order and the selection rules

Theorem 7.1 (Transfer-lattice selection rule)**.** *Grade monomials by energy transfer, $[h_0, M] = [(\Delta n_1)\omega_1 + (\Delta n_2)\omega_2]M$. Order-$n$ objects have transfers in the n-fold sumset of the vertex transfer set, total degree $\leq n + 2$, and degree parity $\equiv n \bmod 2$ (proof: Appendix D). The order-n obstruction is supported on the finite candidate sets: $\{2{:}1\}$ at order 1, $\{3{:}1\}$ at order 2, $\{3{:}2, 2{:}1, 4{:}1\}$ at order 3, with 5:1 first reachable at order 4. The lattice gives candidates; coefficients decide.*

Theorem 7.2 (Third order)**.** *At generic frequencies the third-order obstruction vanishes and R_3 is Hermitian with odd total degrees ≤ 5. At $\omega_1 = \frac{3}{2}\omega_2$ the class is nonzero; at $(\omega_1, \omega_2) = (3, 2)$ its exact value is*

$$\mathfrak{o}_+^{(3)} = -\frac{117\sqrt{30}}{1120}\, i\left(a_1^2 a_2^{\dagger 3} + a_1^{\dagger 2} a_2^3\right), \tag{6}$$

and the coefficient scales exactly as $\omega_2^{-13/2}$ (verified by the ratio $2^{-13/2}$ between $(\omega_1, \omega_2) = (6, 4)$ and $(3, 2)$), so the general form carries the dimensional factor $\omega_2^{-13/2}$. The class is gauge independent in the sense of Theorem 5.1. The candidates 2:1 *(orders 1 and 3) and* 4:1 *(order 3) have vanishing coefficients.*

Conjecture 7.3 (Resonance hierarchy)**.** *The coprime ratio $\omega_1/\omega_2 = p{:}q$ first obstructs at order $p + q - 2$ when the mode-2 transfer p is odd. (Verified:* 3:1 *at order 2,* 3:2 *at order 3, and* 5:1 *at order 4 — Proposition 7.4; externally reproduced exact instances:* 5:3 *and* 7:1 *at order 6 — Proposition 7.5. The mechanism excluding even mode-2 transfers and the all-coprime statement remain open.) The resonance set is an expanding hierarchy, not (so far) a dense set.*

Computational Proposition 7.4 (The 5:1 obstruction at fourth order)**.** *At $(\omega_1, \omega_2) = (5, 1)$ the order-2 and order-3 kernel projections vanish (the selection rule makes* 5:1 *unreachable below order 4) and R_2, R_3 exist and are Hermitian. The fourth-order class is nonzero, exactly*

$$\mathfrak{o}_+^{(4)} = -\frac{203125\sqrt{5}}{2341011456}\left(a_1 a_2^{\dagger 5} - a_1^\dagger a_2^5\right) = -\frac{13 \cdot 5^{13/2}}{2^{16}\, 3^6\, 7^2}\left(a_1 a_2^{\dagger 5} - a_1^\dagger a_2^5\right), \tag{7}$$

the on-shell $1 \leftrightarrow 5$ conversion on the predicted monomial, gauge independent in the sense of Theorem 5.1. The coefficient scales exactly as ω_2^{-9} (ratio 2^{-9} between $(10, 2)$ and $(5, 1)$), extending the dimensional series $\mathfrak{o}_+^{(n)} \propto \omega_2^{-(5n-2)/2}$ for $n = 2, 3, 4$; and $R_4 = O(\epsilon^{-6})$ toward the Jordan boundary, the fourth data point of $R_n = O(\epsilon^{-3n/2})$ (Section 8). The order-n sources are generated programmatically from the adjoint series of Appendix A, and the same code path re-derives the order-2 and order-3 classes exactly before running order 4. (Checks FO1–FO9, `verify_51_order4.py`*.)*

Computational Proposition 7.5 (The 5:3 and 7:1 obstructions at sixth order)**.** *At $(\omega_1, \omega_2) = (5, 3)$ and $(7, 1)$, the exact kernel projections at orders two through five vanish and the corresponding $R_2, \ldots, R_5$ are Hermitian. The first nonzero classes occur at order six and are*

$$\mathfrak{o}_{+,5:3}^{(6)} = \frac{3863828151875\sqrt{15}}{6463101113204736}\left(a_1^3 a_2^{\dagger 5} - a_1^{\dagger 3} a_2^5\right), \tag{8}$$

$$\mathfrak{o}_{+,7:1}^{(6)} = \frac{28633766567\sqrt{7}}{2656254925209600000}\left(a_1 a_2^{\dagger 7} - a_1^\dagger a_2^7\right). \tag{9}$$

They are the predicted $3 \leftrightarrow 5$ and $1 \leftrightarrow 7$ on-shell conversion kernels, with the real coefficient and antisymmetric combination required by even-order parity. Rescaling $(5, 3)$ to $(10, 6)$ multiplies the result by exactly 2^{-14}, matching the sampled dimensional law $\mathfrak{o}_+^{(n)} \propto \omega_2^{-(5n-2)/2}$ at $n = 6$.

The evidence consists of a 31-check exact Forge battery on two code generators, an independent end-to-end Hermiticity assembly through order five, and a from-scratch SymPy recomputation agreeing term by term. The bounded import certificate pins the source commit and artifacts. This proposition does not prove the hierarchy for every coprime ratio, convergence, *or nonremovability under every further admissible canonical transformation. The upstream dirty-toolchain marker is retained for provenance and independently discharged by the clean pinned replay recorded with the import.*

8 The Jordan divergence hierarchy

Theorem 8.1 (Exact first-order Jordan asymptotics)**.** *With $\omega_{1,2} = \omega \pm \epsilon$, the explicit R_1 of (2) has the exact leading Laurent behaviour*

$$R_1 = \epsilon^{-3/2}\,\mathcal{W}\left[\frac{xyp}{2\sqrt{\omega}} + \frac{y^2 q}{4\sqrt{\omega}} - \frac{p^2 q}{4\sqrt{\omega}} - \frac{x^2 q}{2\omega^{5/2}} + \frac{q^3}{6\omega^{5/2}}\right] + O(\epsilon^{-1/2}), \tag{10}$$

so $R_1 = \Theta(\epsilon^{-3/2})$, while the free generator diverges only logarithmically.

Computational Proposition 8.2 (Higher orders)**.** *The solved R_2 and R_3 scale as ϵ^{-3} and $\epsilon^{-9/2}$ (measured decade exponents -3.00 and -4.49 over $\epsilon = 10^{-2} \to 10^{-3}$), with no small-denominator enhancement from $\omega_1 - \omega_2$ at second order. These are high-precision numerical statements at fixed orders, not proved big-O estimates.*

Conjecture 8.3. *$R_n = \Theta(\epsilon^{-3n/2})$, so the deformation's radius of convergence in ζ collapses as $\zeta_c \sim \epsilon^{3/2}$ toward the Jordan boundary.*

9 The perfect-square vertex

The interacting theory is $S = -\frac{1}{2}\int(\Box\phi + \lambda(\partial\phi)^2)^2$ [4], with cubic vertex $V_3 \propto \Box\phi(\partial\phi)^2$ and quartic $V_4 \propto \frac{1}{2}(\partial\phi)^4$.

Proposition 9.1 (Källén form)**.** *On $p_1 + p_2 + p_3 = 0$ the symmetrized cubic vertex is exactly*

$$\mathcal{V}_3 = p_1^2(p_2\cdot p_3) + p_2^2(p_1\cdot p_3) + p_3^2(p_1\cdot p_2) = \tfrac{1}{2}\lambda_{\mathrm{K}}(p_1^2, p_2^2, p_3^2). \tag{11}$$

The polynomial vertex factor therefore has a second-order zero at the massless confluent point*: $\lambda_{\mathrm{K}}(0,0,0) = 0$ and $\nabla\lambda_{\mathrm{K}}(0,0,0) = 0$, while the Hessian is nonzero. On the regulated decay channel, $\lambda_{\mathrm{K}}(m_1^2, m_2^2, m_2^2) = m_1^2(m_1 - 2m_2)(m_1 + 2m_2)$: the vertex vanishes exactly at threshold and turns on continuously above it. (A full external-state selection rule for generalized Jordan legs would additionally require the confluent limit of leg normalizations, shell measures, and mode factors; we do not claim it here.)*

10 First order in the regulated field theory

Regulate by $P_\delta = (\Box + m_1^2)(\Box + m_2^2)$, $\delta = m_1^2 - m_2^2 > 0$, at finite volume (Remark 2.2). In the mode expansion every leg is on its branch shell, so on the kernel the vertex coefficient is $\frac{1}{2}\lambda_{\mathrm{K}}$ of the branch masses. The transported field mode, derived from the free Dyson transport and verified against the spectral Wightman function of [13] (Appendix E), is

$$\phi'(k) = \frac{i}{\sqrt{2\delta}}\left[\frac{a_2 + a_2^\dagger}{\sqrt{\omega_2}} + \frac{a_1 - a_1^\dagger}{\sqrt{\omega_1}}\right], \qquad \sigma = \sqrt{\delta}\ (k\text{-independent}). \tag{12}$$

Theorem 10.1 (First-order protection; sectorwise parity obstruction). *(i)* Even-ghost selection rule*: the transported reality factors make $v^\dagger - v$ vanish on every odd-ghost-count monomial (Appendix E). Even-ghost shells are kinematically closed for all $m_1 > m_2 > 0$. Hence $\mathfrak{o}_+^{(1)} = 0$ identically — below and above the decay threshold. (ii) The sectorwise branch parity $\kappa_0 = (-1)^{N_{\text{ghost}}}$ has $\mathfrak{o}_K^{(1)} = 0$ below threshold and, above it, a nonzero class on the open $1 \to 22$ shell, equal there to $\lambda_{\mathrm{K}}(m_1^2, m_2^2, m_2^2)$ times the leg normalizations.*

Computational Proposition 10.2 (Split-basis scaling). *At massive confluence both R_1 and K_1 diverge as $\delta^{-3/2}$; on massless Jordan paths $m_2^2 = \alpha\delta$ the positive R_1 scales as $\delta^{-1/2}$ with path-independent power, while the Krein K_1 is $\delta^{-1/2}$ generically and $\delta^{-3/2}$ on the exceptional path $\alpha = 2/7$. These are statements about coefficients in the split mode basis, which itself degenerates at $\delta = 0$; a basis-independent formulation on confluent states is deferred.*

11 The exact two-field rewriting

The two-field form below is, up to normalization and sign conventions, the $O(1,1)$ embedding of Bateman and Turok [4], who introduced the pair (Ω, Υ), identified the full $O(1,1)$ symmetry including the exchange as ghost parity and internal charge conjugation, and constructed the quantum embedding relating the two theories. The theorem restates their embedding in our conventions; the content added in this section and the next two is the chart-transition description of the exchange, the null-Jordan-sector and interaction-generated Jordan statements, the location of the sectorwise obstruction inside this structure, and the confluent-parity bridge.

Theorem 11.1 (Two-field form and exchange symmetry [4, adapted]). *Introducing the auxiliary field ψ and the change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ (unit pointwise Jacobian),*

$$\mathcal{L} = -\partial_\mu U\,\partial^\mu V + \frac{\lambda^2}{2}U^2V^2 = -\tfrac{1}{2}(\partial X)^2 + \tfrac{1}{2}(\partial Y)^2 + \frac{\lambda^2}{8}(X^2 - Y^2)^2, \qquad X, Y = \tfrac{U \pm V}{\sqrt{2}}. \tag{13}$$

The exchange $U \leftrightarrow V$ is an exact symmetry of the extended two-field theory. *In the original variables it is the transition map between the charts $\phi_A = \frac{1}{\lambda}\log U$ and $\phi_B = \frac{1}{\lambda}\log V$,*

$$(\phi, \psi) \;\longmapsto\; \Big(\tfrac{1}{\lambda}\log\tfrac{\psi}{\lambda} - \phi,\; \psi\Big), \tag{14}$$

defined on the chart overlap ($U \neq 0 \neq V$), singular at the perturbative vacuum $\psi = 0$, and requiring a branch choice in the complexified theory. The original (ϕ, ψ) chart covers only $U \neq 0$ and does not contain the mirror background: the exchange is a global symmetry of the extension, not an everywhere-defined involution of the original field space, and it is invisible at any finite perturbative order around the original vacuum.

Bateman and Turok warn, and we adopt the caveat, that the two theories are inequivalent beyond perturbation theory: the fourth-order path integral corresponds to $U > 0$, while the two-field model integrates over all U and has an indefinite kinetic form [4]. Statements about the mirror background $(0, 1)$ are therefore statements about the extended theory; whether both null branches are physical sectors of one completion is open.

Proposition 11.2 (Null sectors and interaction-generated Jordan dynamics). *The constant zero-action stationary backgrounds form the two null branches $UV = 0$ (whether they are* *vacua in a given completion is a separate question, since the kinetic form is indefinite). The perturbative background* $(U,V)=(1,0)$ *is a nonzero null configuration; the exchange maps it to* $(0,1)$*. Expanding about* $(1,0)$*:* $\mathcal{L}^{(2)}=-\partial u\partial v+\frac{g}{2}v^2$, $\mathcal{L}^{(3)}=g\,uv^2$, $\mathcal{L}^{(4)}=\frac{g}{2}u^2v^2$ $(g=\lambda^2)$*, with equations* $\Box v=0$, $\Box u=gv$*: the exact* $\Box^2$ *Jordan pair, the off-diagonal supplied by the interaction coupling; about* $(0,1)$ *the mirror expansion has the oppositely oriented chain. A chain-preserving involution commuting with a single Jordan block is* $\pm I$ *(companion lemma), so exchange between oppositely oriented sectors is the only possible realization of a parity-like symmetry. In this frame the interactions are pure potentials:* $H_{\rm int}=-L_{\rm int}$ *exactly, with no Ostrogradsky–Legendre corrections.*

12 The sectorwise second-order obstruction

Regulate the sector-A theory by $\mathcal{L}_0=-\partial u\partial v-\mu^2uv+\frac{g}{2}v^2+\frac{\epsilon}{2}u^2$, with masses $m_\pm^2=\mu^2\pm\sqrt{\epsilon g}$ and branch eigenvectors $v=\rho_b u$, $\rho_\pm=\mp\delta/(2g)$, $\delta=m_+^2-m_-^2$ (the regulator necessarily breaks the exchange symmetry). All computations below set $g=1$. The positive-frame quantization and the complete normalization conventions — finite volume with unit-normalized discrete modes $[a,a^\dagger]=1$, unit-norm external states, lattice momenta, $m_L=4$ — are collected in Appendix E. The second-order on-shell source element is computed from the exact matrix-element form

$$\langle\mathrm{out}|\,\mathcal{S}_+^{(2)}\,|\mathrm{in}\rangle=\langle v_2^\dagger-v_2\rangle+\sum_n\frac{\langle\mathrm{out}|v_1^\dagger|n\rangle\langle n|v_1^\dagger|\mathrm{in}\rangle-\langle\mathrm{out}|v_1|n\rangle\langle n|v_1|\mathrm{in}\rangle}{E-E_n},\tag{15}$$

derived in Appendix E from $(v^\dagger-v)(v+v^\dagger)+(v+v^\dagger)(v^\dagger-v)=2(v^\dagger v^\dagger-vv)$.

Theorem 12.1 (Sectorwise obstruction on an open shell subset)**.** *Consider the branch-changing scattering* $H+L\to L+L$*, kinematically open at suitable momenta for every* $m_H>m_L$*. At the rational center-of-mass point* $m_H=\frac{3}{2}m_L$ *(both incoming particles at rest; outgoing momenta* $\pm\frac{3}{4}m_L$*; in lattice units* $m_L=4$*,* $m_H=6$*, momenta* ±3*, all shell conditions exact), the source element* (15) *has the exact value*

$$\boxed{\mathcal{M}_{\rm obs}=\frac{401\sqrt{6}}{39424\,g^2}\xrightarrow{g=1}\frac{401\sqrt{6}}{39424}}\tag{16}$$

(the $1/g^2$ *dependence at fixed masses is verified exactly: each vertex carries* $g\rho^2\propto\delta^2/g$*), with the quartic contact piece exactly zero. The computation is truncation-complete (Lemma 12.2) and the element is analytic in the shell data on the nonsingular part of the shell; hence* the second-order obstruction is nonzero on a nonempty open subset of the shell containing this point*. Independent tuned kinematic points give* 0.5233, 0.3109, 0.3970 *(numerical), consistent with nonvanishing on a large region; a characterization of the zero set is left open, and no claim is made for every mass pair.*

Lemma 12.2 (Truncation completeness)**.** *For external momenta* $\{k_a,k_b\}\to\{k_c,k_d\}$ *pairwise distinct, every intermediate Fock state contributing to* (15) *has all momenta in the finite set generated by the externals and their pairwise sums and differences (*$s=k_a+k_b$*,* $t=k_a-k_c$*,* $u=k_a-k_d$*): in each contributing chain the surviving spectator particles must coincide with external particles, which fixes every internal momentum; free-momentum loops arise only in* *forward/disconnected kinematics, excluded here. (Proof: Appendix E; corroborated by exact cutoff independence between $K_{\max} = 4$ and 6.)*

Remark 12.3 (Oscillator versus field theory).

PU oscillator	field theory
discrete energy spectrum	continuous momentum spectrum
isolated rational resonance	generic scattering shell
3:1 first appears at order 2	$H + L \to L + L$ appears at order 2
finite resonant multiplet	continuum of degenerate in/out states

The continuum turns the discrete resonance obstruction into a scattering-shell obstruction. The exact exchange symmetry does not cure it: it maps the obstructed sector-A construction to the mirror sector-B construction.

Computational Proposition 12.4 (Krein consistency of the obstructed element). *Let $G = (-1)^{N_{\text{ghost}}}$ be ghost-branch number parity in the positive-frame Fock space, and let ι be the mode identification $(b,k)_B \simeq (b,k)_A$ of the mirror sector. At the rational kinematic point of Theorem 12.1, exactly: (i) $H_B = \iota H_A \iota^{-1}$ (the mirror sector is the same theory in the mirrored frame) and $H_A^\dagger = G H_A G$ term by term, so the mirror-adjoint relation $H_B = W H_A^\dagger W^\dagger$ holds with $W = \iota \circ G$ — i.e. the "doubled pairing" $\eta_{\text{dbl}} = \text{offdiag}(W^\dagger, W)$ is precisely the two-sector unfolding of the ghost-parity Krein metric of [4], not additional structure. (ii) On the degenerate shell $\{|H(0)L(0)\rangle, |L(3)L(-3)\rangle\}$ (exhaustively the whole reachable shell) the second-order on-shell T satisfies $GTG = T^\dagger$ exactly, while Hilbert Hermiticity fails: the parity-even (diagonal) block is real, and the entire obstruction is the parity-odd block, $T(\text{out},\text{in}) = -401\sqrt{6}/78848 = -\mathcal{M}_{\text{obs}}/2$. (iii) Consequently the finite-time interaction-picture S obeys $S_B^\dagger W S_A = W$ identically — Krein pseudo-unitarity is exact on the same matrix elements on which every positive pointed metric is obstructed. A positive H-invariant graph subspace of the doubled space exists if and only if a positive pointed metric exists (standard J-self-adjoint graph argument [10], instantiated and machine-verified), so doubling preserves pseudo-unitarity but cannot evade the obstruction. (Checks DQ1–DQ9, `verify_doubled_theory.py`.)*

The obstruction is therefore carried exactly by the κ-odd component of the on-shell T — the component that the weak-ghost-symmetry positivity mechanism of [4] does not use. Whether it maps into their null orthogonal (C) component under their asymptotic embedding is the natural next question (Section 14).

13 The confluent parity theorem

Theorem 13.1 (Confluent limit of the regulated parity). *Let $P_\delta b_\pm = \pm b_\pm$ be the regulated branch parity and $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$ the confluent Jordan basis, so $P_\delta c = \frac{\delta}{2} d$ and $P_\delta d = \frac{2}{\delta} c$; in the standard column convention on the ordered basis (c, d),*

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \qquad P_\delta^2 = 1, \tag{17}$$

with no bounded limit as $\delta \to 0$: the regulated parity cannot converge as a chain-preserving parity inside one Jordan sector. On the doubled space with the oppositely oriented *confluent identification in the mirror sector ($c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$), the cross parity*

$b^A_\pm \leftrightarrow \pm b^B_\pm$ *equals, exactly and* δ*-independently, the sector exchange* $(c_A, d_A) \leftrightarrow (c_B, d_B)$*. The regulated branch parity thus converges, after doubling by the oppositely oriented Jordan sector, to* the linearization of the exact sector-exchange involution around the two mirror backgrounds. *(Identifying the full nonlinear quantum operator requires a separate construction, not supplied here.)*

14 Discussion

The mechanism. Away from resonances the positive pseudo-Hermitian metric deforms locally, with explicit generators. The failures are structural: on-shell branch-changing processes produce gauge-independent cohomological obstructions — isolated rational loci for the oscillator, generic scattering shells in the field theory; at the oscillator resonances the obstruction is certified spectral; and independent of resonances, the deformation loses uniformity toward Jordan degeneration. The sectorwise branch parity is broken by real ghost decay above threshold; what survives at the massless perfect-square boundary is the exact sector-exchanging involution of the extended two-field theory, whose linearization is the bounded confluent limit of the regulated parity on the doubled space.

Symmetry enhancement at the boundary. The massless perfect-square point is not the split theory with parameters set to zero: the Källén vertex degenerates there, the doubled $O(1,1)$ structure emerges, and the Jordan dynamics is generated by the interaction about its own null backgrounds — a boundary-emergent structure mirroring the Weyl enhancement at the conformal boundary of the gravity companion [14].

Separation of the two completions. Proposition 12.4 sharpens the free-theory picture of the companion series: in the interacting theory the two real forms are not merely distinguished at Jordan degenerations — they *separate*. The doubled Krein pairing is exactly preserved at the split rational shell ($GTG = T^\dagger$ on shell, finite-time pseudo-unitarity), while every analytic positive pseudo-Hermitian completion of the pointed sector is obstructed on an open shell subset. This already excludes identifying that Krein pairing with an ordinary positive pseudo-Hermitian completion. It does *not* establish the stronger Bateman–Turok weak-ghost-symmetry positivity mechanism at split mass: Proposition 14.2 shows that the regulated vacuum image contains both charge signs there. The sharpest remaining form of the reconciliation is therefore a boundary transport question: under their asymptotic embedding, physical operators decompose into a ghost-symmetric part and a null orthogonal part, and only the former contributes to their generalized Born rule [4]; since the obstruction is κ-odd (Proposition 12.4), the conjecture is that it is carried into the null component — obstructing every positive metric while remaining invisible to the Krein probabilities. Verifying this on the computed $H + L \to L + L$ element is the concrete next calculation; the lemma and the proposition below supply its verified ingredients.

The charge-null mechanism. Throughout this paragraph $\dagger$ denotes the Krein adjoint and τ the Krein trace. The following finite-particle lemma makes the null-relocation mechanism self-contained (it is the statement used in [4], whose general proof is deferred to their companion paper; for boost weights it is a three-line grading argument).

Lemma 14.1 (Charge null)**.** *Let a Krein space carry an $SO^+(1,1)$ charge grading whose pairing couples charge q only to charge $-q$, and suppose the Krein adjoint preserves charge — as it does for boost weights, where* c *and* c† *of one field carry the* same *charge. Then $\tau(A_q) = 0$ for every $q \neq 0$. Consequently an operator $C = \sum_{q<0} C_q$ containing only strictly negative charges satisfies $\tau(C^\dagger C) = 0$ and $\tau(B^\dagger C) = 0$ for every neutral B: C is null and orthogonal to the neutral sector, and the generalized Born trace of a weakly ghost-symmetric process depends on its neutral part alone.*

Proof. Boost invariance of the trace gives $\tau(A_q) = \tau(\alpha_\sigma A_q) = e^{q\sigma}\tau(A_q)$ for all σ, hence $\tau(A_q) = 0$ for $q \neq 0$; equivalently, on a charge-graded Fock space a grading-raising operator is traceless. Charge preservation under the adjoint gives $q(C_q^\dagger C_r) = q + r < 0$ and $q(B^\dagger C_r) = r < 0$ term by term. ☐

Computational Proposition 14.2 (Regulated embedding and its charge structure)**.** *At the rational kinematic point, per momentum sector, the regulated split theory admits a canonical Bogoliubov map onto the cross-paired charge basis of a neutral reference theory $-\partial u\, \partial v - \tilde{\mu}^2 uv$ (operators c_U, c_V with the only nonvanishing commutators $[c_U, c_V^\dagger] = [c_V, c_U^\dagger]$; charges $q(c_U^{(\dagger)}) = +1$, $q(c_V^{(\dagger)}) = -1$, the analogue of Bateman–Turok's b_Ω, b_Υ). The pointed vacuum maps to a Gaussian state whose squeezing components obey, exactly and for every reference dispersion,*

$$\frac{S_{UU}}{S_{VV}} = \left(\frac{\delta}{2g}\right)^2 = \frac{\epsilon}{g}, \qquad \delta^2 = 4\epsilon g, \tag{18}$$

i.e. the charge $+2$ component is sourced by the regulator $\frac{\epsilon}{2}u^2$ and the charge -2 component by the interaction-generated $\frac{g}{2}v^2$. The embedding is strictly one-sided — no positive charges in the vacuum image, the hypothesis of Lemma 14.1 for the mapped theory — if and only if $\epsilon = 0$: the $O(1,1)$-symmetric confluent line. The massless Bateman–Turok theory is the point $\epsilon = \mu^2 = 0$ on that line. At confluence the matched-reference limit is $S_{VV} \to -g/(4w^2)$, or $-1/(4w^2)$ in the $g = 1$ normalization used in the computation; at $\mu^2 = 0$, where $w = |\mathbf{k}|$, this is precisely the coefficient structure of their Eqs. (C5)–(C6). At split masses the exact ratio above rules out removing the $+2$ component by changing the reference dispersion. Under the residual charge-preserving Bogoliubov freedom, numerical solution of the adaptation equations runs to a degenerate (confluent-type) frame rather than finding a bounded adapted frame. (Checks ON1–ON4, `verify_obstruction_null.py`*.)*

The proposition is the charge-frame image of the first-order result of Section 10: real ghost decay above threshold breaks the sectorwise branch parity, and in the charge frame the same ϵ sources the charge $+2$ contamination that violates weak ghost symmetry. The relocation of the κ-odd obstruction into a null negative-charge component is therefore exact on the one-sided confluent line, with the Bateman–Turok identification at its massless point, and holds at split masses with charge contamination of relative size ϵ/g (not assumed small away from the boundary). The remaining, precisely-posed step — the obstruction-to-null theorem — is the boundary Born-trace evaluation: the neutral component's blindness to the obstruction coefficient and its limit along a specified on-shell family with $(\epsilon, \mu^2) \to (0, 0)$. One exact rational family is $m_L = 4s$, $m_H = 6s$, $|\mathbf{k}_{\text{out}}| = 3s$, $\mu^2 = 26s^2$, $\epsilon g = 100s^4$, followed by $s \to 0$; the process operator must be transported along this family rather than held fixed while only the embedding is varied. Lemma 14.1 makes that step self-contained.

That the transport is essential, rather than a technical scruple, can be quantified. Evaluating the generalized Born trace of [4] directly on the two-dimensional shell of Proposition 12.4, with the process operator held fixed at the split-mass rational point, gives $\tau(T^\dagger T) = \|T_+\|^2 - \|T_-\|^2$ — the κ-graded components being orthogonal in the shell trace — with

$$\|T_-\|^2 = 2\Big(\frac{401\sqrt{6}}{78848}\Big)^2 = \frac{482403}{1554251776} \neq 0. \tag{19}$$

Lemma 14.1 sends exactly this quantity to zero. A truncated shell carries no boost action, so the lemma's hypothesis is absent and its conclusion fails by precisely the obstruction squared; the fixed-shell trace is nonetheless positive, but only because $\|T_+\|^2$ exceeds $\|T_-\|^2$ by a factor of about 26, which is a statement about the relative size of two matrix elements and not about nullity. A fixed-shell evaluation therefore cannot decide the obstruction-to-null theorem in either direction. We record it because the positive sign is easy to mistake for confirmation. (Machine-checked in `REVERSE_PHYSICS_BT_BORN_TRACE_V1`; the exact value of $\|T_+\|^2$ and the positivity witness are given there, and the same certificate repairs a floating-point contamination of the diagonal of T in `verify_doubled_theory.py` which affected no published quantity, every such quantity being an off-diagonal element or a reality statement.)

One structural simplification is worth recording alongside it: on a finite shell the weak ghost symmetry of [4] — existence of a decomposition $A = B + C$ with B ghost symmetric and C null and orthogonal to B — is *equivalent* to the single inequality $\|A_-\| \leq \|A_+\|$, since the κ-eigenspaces are orthogonal in the shell trace and $\tau(A^\dagger A) = \|A_+\|^2 - \|A_-\|^2$. There is nothing to search for. This does not extend to the charge-graded trace, where Lemma 14.1 rather than a norm comparison carries the positivity.

Complementarity of the two premises. The separation above is easily read as a scoreboard, and it is not one. Each completion rests on a premise that fails somewhere the other's holds, and the two failures occur on *different* boundaries. The positive pseudo-Hermitian completion requires the dynamics to be diagonalizable with real spectrum; that fails precisely on the coincident-pole locus, where the similarity transformation is singular and only the Krein real form continues as a nondegenerate completion [14]. The Krein completion requires the internal charge to be one-sided; Proposition 14.2 shows this holds exactly iff $\epsilon = 0$, and at split masses both signs are present. Neither premise is generic. They are boundary conditions on different boundaries, so neither completion is a specialization of the other and neither is discardable in favour of it.

Two consequences follow, and the second is the one that bears on conformal gravity. First, the free-level agreement of [14] is not a weaker form of the interacting separation: it says that any dispute between the two completions conducted at free-field level concerns the involution and not the predictions, since both induce the same functional on the gauge-invariant algebra. Second, at loop order neither completion currently reaches the Jordan locus — which is where a pure $1/k^4$ propagator sits. The positive pseudo-Hermitian route withdraws the standard cutting rules there by its own account ([3], Sec. VI; the argument is set out in [14]), while [4] establish positivity at tree level and name collinear infrared divergence of the massless theory as the obstacle to higher orders. The two limitations are unrelated in origin and coincide in location.

That scalar limitation can now be sharpened on one complete declared carrier.

Remark 14.3 (Axis-compatible collinear normalization obstruction). At order λ^6, on the complete final-state collinear logarithmic normalization carrier, differential in the nonsingular outer two-body solid angle, the three identical real-emission pair boundaries respond to a common daughter mass-ratio normalization $r \mapsto cr$ as

$$\Delta_c \frac{d\sigma_{\text{real}}}{d\Omega} = \frac{3\lambda^6}{512\pi^4 s} \log c. \tag{20}$$

Let the recombined virtual parent regulator be $G(x, y) = xg(y/x)$, with g continuous at zero and $0 < |g(0)| < \infty$, while the spectator regulators are held fixed. Then the complete virtual external-mass logarithmic response is zero at constant order. Hence the real and virtual normalization responses do not cancel on this axis-compatible class. The statement has dependency tag `REDUCED-MODE`; it is not a complete NLO probability or a theorem against distributional, dressed-state, enlarged-degenerate-state, or resummed constructions.

Certificate summary. For $x_i = \delta a_i$ and pair invariant $t = \delta\tau$, the exact three-spectator square-free jet of the complete 25-graph tree gives

$$[a_2a_3a_4]C^2 = \frac{3(a_0 - a_1)^2\big((a_0 - a_1)^2 - 2\tau(a_0 + a_1)\big)}{8\tau^3}, \qquad A_5 = \frac{M_5}{8\lambda^3} = \delta^2 C + O(\delta^3). \tag{21}$$

The splitting fraction and outer scattering ratio cancel identically, so the inner solid angle contributes 4π. The exact threshold finite part shifts by $-(3/8)\log c$. Restoring the five-delta-prime sign $(-1)^5$, the $1/(2!3!)$ projector weight, factorized three-body phase space, and $|M_5|^2 = 64\lambda^6 A_5^2$ gives $\lambda^6 \log c/(512\pi^4 s)$ per unordered pair and therefore (20). The certified virtual boundary rate is

$$\frac{d\sigma_{\text{virt,boundary}}}{d\Omega} = \frac{3\lambda^6}{128\pi^4 s} \sum_{i=1}^{4} \log \frac{-\mu^2}{X_i}. \tag{22}$$

For an axis-compatible parent, $G(x, cy)/G(x, y) = g(cr)/g(r) \to 1$ as $r = y/x \to 0$; its logarithm therefore has zero constant response. The hard Mandelstam-ratio logarithm also has zero daughter-ratio response, while continuous cut-free terms and local analytic counterterms cannot supply one. Both the dot-product graph producer and a method-distinct invariant-Källén verifier establish these identities exactly in the axis-gluing certificate cited above.

Remark 14.4 (Finite coherent collinear projector transport). Let $B = 3\lambda^4/(32\pi^2 s)$ be the Born rate and set $\eta = \lambda^2 \log c/\pi^2$. The absolute real response $\lambda^6 \log c/(512\pi^4 s)$ per pair is therefore $B\eta/48$, not a dimensionless $1/512$ projector coefficient. Place one hard channel and the three unordered collinear-pair channels in the carrier $(h, r_{12}, r_{13}, r_{23})$ and let

$$P_0 = \operatorname{diag}(1, 0, 0, 0), \qquad K = \begin{pmatrix} 0 & -a & -a & -a \\ a & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ a & 0 & 0 & 0 \end{pmatrix}, \qquad a = \frac{\sqrt{3}}{12}.$$

Then $K^T = -K$, and the exact formal transport $P(\epsilon) = e^{\epsilon K} P_0 e^{-\epsilon K}$ obeys $P(\epsilon)^2 = P(\epsilon)$ and $\operatorname{tr} P(\epsilon) = 1$ through order two. Its order-two diagonal response is

$$(P_2)_{hh} = -3a^2 = -\frac{1}{16}, \qquad \sum_{r=1}^{3} (P_2)_{rr} = 3a^2 = \frac{1}{16}.$$

More generally, if the order-one hard–collinear block is A, the equation $P^2 = P$ uniquely forces $(P_2)_{hh} = -AA^T$ and $(P_2)_{cc} = A^T A$; the displayed negative term is therefore normalization, not an adjustable weight. Multiplication by $B\eta$ gives the physical hard response $-3\lambda^6 \log c/(512\pi^4 s)$ and cancels all three real pairs.

There is nevertheless an exact dynamical obstruction to the ordinary dressing ansatz. For massless collinear daughters the published cubic vertex has $\mathcal{M}_3 = -i\lambda t^2$, whereas $\Delta E = t/(2E) + O(t^2)$. Hence the ordinary Dyson/Fock kernel $\mathcal{M}_3/\Delta E$ vanishes as $t \to 0$ and its Gram cannot be $1/48$. Direct expansion of the published composite map now sharpens this statement. As printed, its Appendix C mode expansion cannot imply both subsequent oscillator formulas; exchanging the ordinary and growing labels in that mode expansion repairs both. With that declared repair the complete resonant two-annihilator order-λ coefficient follows from $R^\dagger \Omega R = \lambda^{-1} e^{\lambda\phi}$ and $R^\dagger \Upsilon R = \lambda^{-1} e^{-2\lambda\phi} \Box e^{\lambda\phi}$. Although its intermediate a-basis entries contain t and t^2, every such term cancels after inversion onto the (Ω, Υ) carrier. A formal finite-wave-packet anti-Krein cubic generator therefore exists without fitting its norm.

This does not yet give the displayed $1/48$. Contracting the ordered daughter slots of the derived kernel with the off-diagonal BT one-particle form, before Bose and phase-space factors, produces a cross Gram proportional to

$$-\frac{E^2(e_1^2 + e_2^2 - e_1 e_2)}{2e_1^3 e_2^3}, \qquad E = e_1 + e_2,$$

which has cubic poles at both splitting endpoints. The continuum distributional extension, oscillatory creation and vacuum-squeeze sectors, background charge bookkeeping, complete incoming projector, and physical NLO probability remain unconstructed. The bare broken-vacuum cubic has charge -1, so neutrality also cannot be inferred before that missing bookkeeping is supplied.

The endpoint extension is itself now classified. The dimensionless cross shape is exactly $-\frac{1}{2}[z^{-3} + (1-z)^{-3}]$. Reflection-even scaling-degree-three extensions retain three independent endpoint distributions. In particular, the triple-plus prescription evaluates to zero on the inclusive constant test, while the symmetric cutoff finite part evaluates to $1/2$; adding $c_0(\delta_0+\delta_1)$ shifts that value by $2c_0$. Thus $c_0 = 1/96$ would reproduce $1/48$, but only by fitting it. The omitted oscillatory and squeezed sectors must dynamically fix the endpoint constants before the finite projector becomes a prediction.

Charge selection now excludes that last proposed source. The oscillatory $a_1^\dagger$ term maps to $b_\Upsilon^\dagger$ with charge -1, while every term in Q_t has charge -2; the BT dagger preserves these charges and the certified inclusive kernel preserves the strictly negative trace radical. They cannot shift neutral endpoint constants. The apparent coisometric range gap is, however, absent on the formal perturbative branch. The published free cross-CCR gives $[A_\Upsilon, A_\Omega^\dagger] = (2E)^3/(4E^2) = 2E$, while the homomorphism gives the same commutator as $2E\Pi$, $\Pi = R_t^\dagger R_t$; hence $\Pi_0 = 1$. Expanding $\Pi^2 = \Pi$ then forces every positive-order coefficient of $\Pi - 1$ to vanish recursively. Formal inversion of R_t is therefore cleared, and the support projection cannot supply the missing coefficient. The exact certificate `REVERSE_PHYSICS_BT_CANONICAL_ENDPOINT_AMBIGUITY_V1` then tests the remaining algebraic conditions directly. On a carrier with one hard and three reflection-even endpoint directions, every $K(u) = \begin{pmatrix} 0 & -u^T \\ u & 0_3 \end{pmatrix}$ is skew, and $e^{\epsilon K} P_0 e^{-\epsilon K}$ preserves the CCR, idempotence, and trace for arbitrary u. Its order-ϵ^2 hard and endpoint traces are $-u^T u$ and $+u^T u$. Hence the listed canonical identities admit a three-parameter countermodel to uniqueness. The choice $u = (\sqrt{3}/12, 0, 0)$ is compatible with $1/48$, but selecting it would be a fit until the deferred continuum map or an independent physical endpoint condition fixes u. This insufficiency witness does not claim that every u is realized by the BT dynamics.

Remark 14.5 (Physical ultraviolet hard-scattering law). The Born rate, perfect-square beta function, and projected virtual hard log are

$$\frac{d\sigma_{\text{Born}}}{d\Omega}=\frac{3\lambda^4}{32\pi^2 s},\qquad \beta_\lambda=-\frac{5\lambda^3}{16\pi^2},\qquad \frac{d\sigma_{\text{virt,log}}}{d\Omega}=\frac{5\lambda^6}{256\pi^4 s}(L_s+L_t+L_u).$$

At fixed nonforward angle, all three channel invariants scale with s. The exact scale-derivative identity is therefore

$$-4\left(\frac{5}{16}\right)\left(\frac{3}{32}\right)+2\cdot 3\left(\frac{5}{256}\right)=0.$$

Thus the explicitly computed hard logarithm supplies precisely the coefficient required by the independently computed running of the Born rate. With

$$D(s)=\lambda_0^{-2}+\frac{5}{16\pi^2}\log\frac{s}{s_0},\qquad \lambda_0=\lambda(\sqrt{s_0}),$$

the leading ultraviolet logarithms resum to

$$\frac{d\sigma^{\text{LL}}_{\text{hard}}}{d\Omega}=\frac{3}{32\pi^2 sD(s)^2}=\frac{24\pi^2}{25s\log^2(s/\Lambda^2)},\qquad \Lambda^2=s_0e^{-16\pi^2/(5\lambda_0^2)}.$$

For a fixed detector acceptance $\theta_0\le\theta\le\pi-\theta_0$, $0<\theta_0<\pi/2$, this gives

$$\lim_{s\to\infty} s\log^2(s/\Lambda^2)\,\sigma^{\text{LL}}_{\text{window}}=\frac{96\pi^3}{25}\cos\theta_0.$$

The result is positive on the asymptotically-free UV branch, and its leading $1/\log^2 s$ coefficient is invariant under analytic coupling-scheme changes. It is the physical short-distance baseline for a nonforward acceptance, not an infrared-complete exclusive particle-number probability. Real–virtual endpoint cancellation, the Jordan/R_t dressing, incoming degenerate sectors, and next-to-leading-log finite terms remain unconstructed.

Where the double pole goes. One structural remark, since it bears on which route is the likelier to extend. The obstruction that withdraws the cutting rules at coincident poles is that the cut weight there is a derivative of a delta rather than a measure of definite sign. That is a property of the fourth-order *variable* and not of the theory: in the two-field frame of [4] the same dynamics is carried by two second-order fields whose propagator is off-diagonal, so every pole is simple and every cut an ordinary delta. The double pole is removed by the change of variables rather than by any additional hypothesis. That is a plausible reason why the Krein route reaches tree level at the locus where the positive route stops, and it suggests the positive-metric programme could be continued past its own Sec. VI in those variables rather than in the fourth-order field.

Open questions. Whether the two pointed sectors admit parity vacua $(|0_A\rangle \pm |0_B\rangle)/\sqrt{2}$ or are superselected requires finite-volume overlap and zero-mode analysis — in the hyperbolic polar coordinates $r = UV$, $\chi = \frac{1}{2}\log(U/V)$ (a coordinate presentation of the $O(1,1)$ model: boost $=$ χ-shift, exchange $= \chi \mapsto -\chi$, conserved current $j = V\partial U - U\partial V = 2r\,\partial\chi$) this becomes a boundary/self-adjoint-extension problem at the null locus $r = 0$, where the conserved charge produces a singular q^2/r barrier. The quantum implementation of the exchange on a completion, and any probability interpretation beyond [4], remain open; so does a normal-ordered operator Ward identity for the internal charge (the classical identity, with its exact regulator breaking $\epsilon u(1+u) - \mu^2 v$, is verified). The hierarchy conjecture's 5:1 prediction is confirmed at fourth order (Proposition 7.4), and its 5:3 and 7:1 instances are confirmed at sixth order (Proposition 7.5); higher coprime loci and the mechanism excluding even mode-2 transfers remain open. The ordinary scalar independent-mass route is no longer merely deferred: Remark 14.3 supplies its first complete collinear logarithmic obstruction. Remark 14.4 shows that coherent projector transport passes the finite coefficient and charge preflights. Its successor must now derive the splitting generator from the broken-vacuum BT/PS asymptotic Hamiltonian, construct the continuum and incoming degenerate sectors, and only then evaluate the quotient trace. The gravitational extension has meanwhile been carried out at tree level in the sequel: the exact Einstein–Weyl cubic and quartic vertices, physical helicities, and Ward/BRST checks produce a second-order metric obstruction. That result does not import the scalar infrared proposition; any Bateman–Turok-type gravitational completion would need its own parent/daughter and asymptotic-state construction. No CPT claim is made: the exchange is a linear sector map, not a constructed antiunitary CPT operator.

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A The Weyl–Moyal recursion

Operators are represented by Weyl symbols on the two canonical pairs (x, p), (y, q); the star product is $f \star g = \sum_n \frac{1}{n!}(i/2)^n \Lambda^n(f, g)$ with $\Lambda = \overleftarrow{\partial}_x \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_x + \overleftarrow{\partial}_y \overrightarrow{\partial}_q - \overleftarrow{\partial}_q \overrightarrow{\partial}_y$, a finite sum on polynomials. For quadratic h_0, $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$ exactly; for cubic–cubic commutators the Λ^3 (constant) term contributes. Expanding $h_\zeta = e^{-X/2} \star (h_0 + \zeta v) \star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2 + \zeta^3 R_3$, with $e^{-X/2} A e^{X/2} = A + \frac{1}{2}[A, X] + \frac{1}{8}[[A, X], X] + \frac{1}{48}[[[A, X], X], X] + \cdots$, the order-$\zeta^2$ and ζ^3 Hermiticity conditions are

$$[h_0, R_2] = (v_2^\dagger - v_2) + \tfrac{1}{2}[R_1, v + v^\dagger], \tag{23}$$

$$[h_0, R_3] = -(T_3 - T_3^\dagger), \tag{24}$$

$$T_3 = \tfrac{1}{8}\big([[h_0, R_1], R_2] + [[h_0, R_2], R_1]\big) + \tfrac{1}{48}[[[h_0, R_1], R_1], R_1] + \tfrac{1}{2}[v, R_2] + \tfrac{1}{8}[[v, R_1], R_1],$$

using $[h_0, R_1] = v^\dagger - v$ and the fact that $[v - v^\dagger, R_1]$ is Hermitian. Equations $[h_0, R] = S$ are solved monomial by monomial in the mode variables $a_1^m a_1^{\dagger n} a_2^r a_2^{\dagger s}$, where $[h_0, M] = \Omega_M M$ with $\Omega_M = (n - m)\omega_1 + (s - r)\omega_2$: $R_M = S_M/\Omega_M$ off the kernel; the kernel component is the obstruction.

B The 3:1 projection and gauge independence

At $\omega_1 = 3\omega_2$ the second-order source $S_2 = \frac{1}{2}[R_1, v + v^\dagger]$ is a quartic polynomial; its kernel monomials must satisfy $(n-m)3\omega_2 + (s-r)\omega_2 = 0$ with degree ≤ 4, i.e. $(m-n, r-s) = \pm(1,-3)$, realized only by $a_1 a_2^{\dagger 3}$ and $a_1^\dagger a_2^3$. Direct evaluation of the two coefficients gives $\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^4}(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$. Gauge independence: a change $R_1 \to R_1 + G$ with $[G, h_0] = 0$ shifts S_2 by $\frac{1}{2}[G, v+v^\dagger]$; transfer bookkeeping shows that for every Hermitian kernel generator (N_1, N_2 polynomials and the resonant quartics $a_1 a_2^{\dagger 3} + a_1^\dagger a_2^3$, $i(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$) the shifted source has no kernel component: the transfers of $v + v^\dagger$, shifted by $\pm(1,-3)$ or $(0,0)$, never land on $\pm(1,-3)$; machine verification confirms $\Pi_{\ker \mathrm{ad}_{h_0}}[G, v+v^\dagger] = 0$ term by term. Anti-Hermitian generators A_1 enter through $\frac{1}{2}[v - v^\dagger, A_1]$, and the same bookkeeping gives $\Pi_{\ker \mathrm{ad}_{h_0}}[v - v^\dagger, A_1] = 0$. Kernel additions to R_2 drop out of the second-order equation since $[h_0, R_2^{\ker}] = 0$.

C The effective shell matrix

Standard second-order degenerate perturbation theory on the shell $\mathcal{E}_{27} = \mathrm{span}\{|j, 27-3j\rangle\}$ gives $K_{ab}^{(2)} = \sum_{m \notin \mathcal{E}} v_{am} v_{mb}/(27 - E_m)$ (the first-order block vanishes: cubic v cannot realize the transfer $\pm(1,-3)$). Evaluating the sums yields the rational diagonal $D(n_1, n_2) = (633n_1^2 - 1836n_1 n_2 - 285n_1 + 3969n_2^2 + 3051n_2 + 818)/26880$ and the antisymmetric off-diagonal quoted in Theorem 6.2. For a tridiagonal matrix only the products $K_{j,j+1}K_{j+1,j} \in \mathbb{Q}$ enter the characteristic polynomial, computed exactly by the three-term recurrence $p_j = (D_j - \kappa)p_{j-1} - K_{j-1,j}K_{j,j-1}p_{j-2}$; the Sturm count of the resulting degree-10 rational polynomial is exactly 8.

D The transfer lattice

Transfer additivity: the grading by ad_{h_0} satisfies $[h_0, AB] = [h_0, A]B + A[h_0, B]$, so transfers add under operator products, hence under star products and commutators. Degree bound: each commutator of polynomial factors lowers total degree by at least two (Λ^1), so order-n objects, built from n cubic factors and $n-1$ commutators (with $[h_0, \cdot]$ degree-preserving), have degree $\leq 3n - 2(n-1) = n + 2$; Moyal corrections lower degree by even amounts, fixing the parity $\equiv n \bmod 2$. A kernel monomial with transfer (d_1, d_2), $d_1 d_2 < 0$, requires degree $\geq |d_1| + |d_2|$ with matching per-mode parities, giving the candidate sets of Theorem 7.1.

E Field conventions, the source formula, and truncation completeness

Conventions. Finite volume; discrete momenta k (lattice units with $m_L = 4$ in Theorem 12.1); modes (b, k) with $[a_{b,k}, a_{b',k'}^\dagger] = \delta_{bb'}\delta_{kk'}$; unit-norm external states; $g = 1$; deformation parameter ζ multiplying H_3, ζ^2 multiplying H_4. Positive-frame legs from the branch eigenvectors: u-legs $(a_+ + a_+^\dagger)/\sqrt{2\omega_+\delta}$ (healthy) and $(a_- - a_-^\dagger)/\sqrt{2\omega_-\delta}$ (ghost, quarter-turn); v-legs $\rho_\pm$ times u-legs, $\rho_\pm = \mp\delta/(2g)$. Interactions $H_3 = -g\int uv^2$, $H_4 = -\frac{g}{2}\int u^2 v^2$ are pure potentials. The transported fourth-order field of Section 10 is fixed by matching the free two-point function $[e^{-i\omega_1 t}/2\omega_1 - e^{-i\omega_2 t}/2\omega_2]/\delta$ of [13]; the even-ghost rule follows because a monomial with g ghost legs, c_g of them creators, has v-coefficient $\propto i^{\#\text{legs}}(-1)^{c_g}$ and $v^\dagger$-coefficient $\propto$ the conjugate with $c_g \to g - c_g$, whose difference vanishes for odd g.

Source formula. With $E_{\text{in}} = E_{\text{out}} = E$ and R_1 solved off shell, $\langle \text{out}|[R_1, w]|\text{in}\rangle = \sum_n [(v^\dagger - v)_{\text{out},n} w_{n,\text{in}} + w_{\text{out},n}(v^\dagger - v)_{n,\text{in}}]/(E - E_n)$ with $w = v + v^\dagger$; the algebraic identity $(v^\dagger - v)w + w(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$ reduces this to (15).

Truncation completeness (proof of Lemma 12.2). Second-order chains pass through 1-, 3-, or 5-particle intermediates. 1-particle: momentum $k_a + k_b$ fixed. 3-particle: the first vertex annihilates one external and creates a pair $(q,\, k_x - q)$; for the second vertex to reach the 2-particle out-state, the surviving created particle must be an out-particle, forcing $q \in \{k_c, k_d, k_x - k_c, k_x - k_d\}$; annihilating both created particles instead returns the in-momenta, giving zero overlap with a distinct out-state. 5-particle: the created triple (p, q, r), $p+q+r = 0$, must contain both out-particles, fixing the third momentum. Hence all internal momenta lie in the s, t, u-generated set, and enlarging the momentum cutoff cannot change the element — as the exact agreement at $K_{\max} = 4, 6$ confirms.

F The doubled confluent parity

With change-of-basis matrix C whose columns are (c, d) in branch coordinates, $P_\delta^{(c,d)} = C^{-1}\,\text{diag}(1, -1)C = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}$. For the doubled space take $C_A = C$ and the oppositely oriented C_B (columns $c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$); the cross map $\kappa(b_\pm^A) = \pm b_\pm^B$ has matrix $T^{-1}\kappa_{\text{branch}}T$ with $T = \text{diag}(C_A, C_B)$, which evaluates to the δ-independent exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$, squaring to the identity.

G Verification checks and their classification

Check	Supports	Type
ID1–ID5	Theorem 3.1	exact symbolic
ID6–ID8	Theorem 4.1	exact symbolic
ID9a–b	Theorem 5.1	exact symbolic
ID10, O3e	Comp. Prop. 8.2	high-precision numeric
SR1–SR3	Theorem 7.1	exact combinatorial
O3a–O3d	Theorem 7.2	exact symbolic
PT1–PT2	Theorem 6.2 (structure)	exact symbolic
(new) Sturm count	Theorem 6.2 (pair)	exact rational
PT3	Theorem 6.2 (truncated op.)	numeric
PT4	broken-PT tongues	numeric (effective matrix)
PS1–PS2	Proposition 9.1	exact symbolic
PS-A–PS-G	Theorem 10.1, Comp. Prop.	exact symbolic + numeric
PS-H	Jordan-chain lemma	exact symbolic
TF1–TF7	Theorem 11.1, Prop. 11.2	exact symbolic
SO1–SO7	Theorem 12.1 (numerics)	exact + numeric
HX1	Theorem 12.1 (exact value)	exact symbolic
HX2–HX3	Theorem 13.1	exact symbolic
(new) PT proposition	Proposition 6.1	exact symbolic
(new) $\omega_2^{-13/2}$ ratio	Theorem 7.2	exact symbolic
(new) R_1 Laurent	Theorem 8.1	exact symbolic
DQ1–DQ2, DQ9	$O(1,1)$ polar form, current, Ward	exact symbolic
DQ3–DQ6	doubled pairing, graph, pseudo-unitarity	numeric
DQ7–DQ8	Comp. Prop. 12.4	exact symbolic
FO1–FO9	Comp. Prop. 7.4, Conj. 7.3	exact symbolic
ON1–ON4	Lemma 14.1, Comp. Prop. 14.2	exact symbolic + numeric
BT-RV	Remark 14.3	exact symbolic/rational
BT-AG	Remark 14.4	exact rational/algebraic
BT-UV	Remark 14.5	exact rational/RG consistency

Scripts, in `symbolic/`, each printing `ALL PASS`:
`verify_interaction_deformation.py` `verify_interaction_order3.py`
`verify_pt_breaking.py` `verify_perfect_square.py` `verify_two_field.py`
`verify_sector_obstruction.py` `verify_hardening.py`
`verify_doubled_theory.py` `verify_51_order4.py`
`verify_obstruction_null.py`

The scalar real–virtual extension is reproduced independently from the repository root by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_real_virtual_axis_gluing.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_real_virtual_axis_gluing.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_real_virtual_axis_gluing
```

The corrected finite-projector and asymptotic-generator preflight is reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_asymptotic_generator_preflight.py --check
```

```
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_asymptotic_generator_preflight.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_asymptotic_generator_preflight
```

The dynamically derived order-λ R_t kernel, Appendix C label consistency test, secular cancellation, and endpoint Gram are reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_rt_jordan_kernel.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_rt_jordan_kernel.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_rt_jordan_kernel
```

The physical ultraviolet hard/window law is reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_uv_hard_scattering_law.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_uv_hard_scattering_law.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_uv_hard_scattering_law
```